Fair Readmission Prediction with Explicit Models of Missing Clinical Data

ICT 5143: Algorithms Fairness and Graphical Models Laboratory | Problem 2
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Abstract. We test whether explicitly modelling laboratory availability improves fairness robustness when missingness changes. A Bayesian network (BN) with observation nodes is fitted by EM and queried by exact variable elimination on 99,343 hospital encounters from the UCI Diabetes 130-US Hospitals dataset. In the pre-registered comparison (train MAR@30%, test MNAR@30%), marginalisation did not significantly improve worst-group recall over imputation or indicator baselines after Holm correction. Discrimination is weak (AUROC ≈ 0.62), subgroup support is limited, and a pattern-aware threshold rule did not improve the primary fairness gap. Results are therefore a qualified negative finding.

1 Question and study design

We simulate a follow-up service selecting the 20% of discharged diabetic patients with highest predicted 30-day readmission risk; 10% and 30% capacities are robustness checks. $Y = 1$ denotes a *recorded* readmission within 30 days. The pre-registered question is whether explicit availability modelling reduces deterioration in worst-group recall when missingness shifts from MAR@30% in training to MNAR@30% in testing. The study is an audit, not a clinical decision system.

The dataset contains 101,766 encounters. We exclude 2,423 expired/hospice discharges, leaving 99,343 encounters from 69,990 patients ($P(Y = 1) = 0.114$). Race (six recorded categories, with "?" retained as Unknown), gender (three categories), and age band are treated as recorded, not inferred. A1C is unmeasured in 83.1% of encounters; this is represented as missing rather than as a clinical result.

2 Models, fairness and evaluation

Let A denote race, gender and age; C diagnosis group; H prior inpatient visits; P prior outpatient/ED visits; $X = (ND, XA)$ number of diagnoses and A1C; and $R = (RN, RA)$ their observation indicators. The core BN factorisation is

$$p(A)\, p(C, H, P \mid age)\, p(ND \mid age, C)\, p(XA \mid age, C, H)\, p(R \mid A, C, H)\, p(Y \mid age, H, P, C, ND, XA).$$

BN-marg marginalises missing values by exact variable elimination; EM estimates parameters from incomplete records. BN-obs adds $Y \to R$ as a non-ignorable sensitivity model. These are associational models, not causal claims. The graph is shown in Fig. 1.

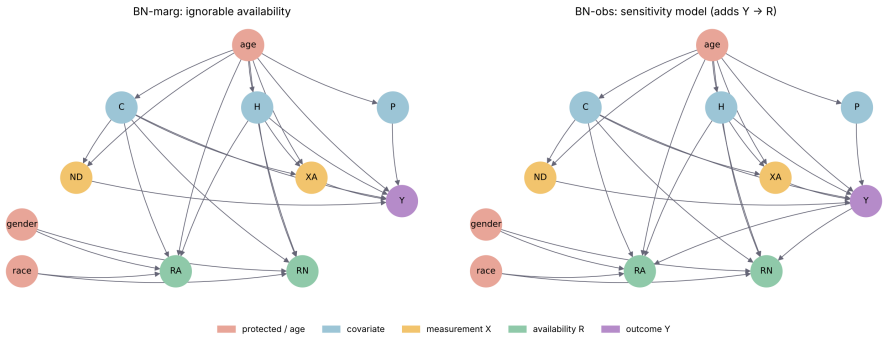


Figure 1: Graphical structures used in the notebook. BN-obs allows readmission to affect observation indicators.

Prediction models do not use race or gender. Baselines include logistic regression and gradient boosting with imputation and/or missingness indicators, complete-case LR, BN-base (mode imputation), BN-marg, and BN-obs variants. A missingness-pattern-aware threshold rule (PAT) adjusts scores using availability patterns only; offsets are validation-tuned under a recall-utility constraint. A group-aware equal-opportunity (EO) baseline uses race×gender at decision time and is reported only as a flagged comparator.

The primary fairness measure is equal-opportunity gap (ΔEOpp) across race×gender cells with at least 30 test positives; worst-group recall is reported alongside it. Utility is recall@20%, with acceptable loss no greater than 0.01 versus LR-ind. Patient-disjoint train/validation/test splits contain 59,814/19,727/19,802 encounters, with no patient overlap. Models are tuned on validation data and frozen. Five seeds vary masking/model randomness; 1,000 patient-cluster bootstrap replicates provide paired intervals. The confirmatory family has three comparisons and Holm correction. Masking scenarios are MCAR, MAR, MNAR and group-dependent, at 10–50% rates.

3 Results

A1C availability is unequal: 16.2% for Caucasian, 18.4% for African American and 25.0% for Hispanic encounters; it is 25.2% for patients under 50 versus 13.5% for those aged 70+. On first encounters, association is strongest for age (Cramér's $V = 0.111$), and smaller for race (0.048) and gender (0.025). Readmission prevalence is 0.099 when A1C is measured and 0.118 when it is not. A conditional test does not reject the specified ignorability check ($\chi^2 = 0.86$, $p = 0.355$); this does not establish MAR.

Table 1: Natural test set, 20% capacity ($n = 19,802$; 2,216 positives). Higher AP/recall/worst-group recall and lower ΔEOpp are preferred.

Method	AP	Recall	Worst recall	ΔEOpp
LR-imp	.1662	.3330	.2744	.0909
LR-ind	.1665	.3326	.2744	.0909
HGB-ind	.1806	.3312	.2744	.0866
BN-base	.1702	.3231	.2977	.0725
BN-marg	.1677	.3294	.2977	.0780
BN-obs	.1672	.3285	.2977	.0725
BN-obs-imp	.1690	.3163	.2791	.0745
LR-ind+EO†	.1665	.3344	.2465	.1458
LR-ind+PAT	.1665	.3384	.2744	.0973
BN-marg+PAT	.1677	.3272	.2977	.0835
Complete-case LR	.1383	.3147	.2059	.2077

†Uses race×gender at decision time. Complete-case results concern a smaller selected subset.

The confirmatory test found no statistically supported advantage for BN-marg under the MAR-to-MNAR shift:

Comparison	Effect*	95% CI	Raw p	Holm p
BN-marg vs LR-imp	−0.0080	[−0.0316, .0148]	.603	1.000
BN-marg vs LR-ind	−0.0019	[−0.0283, .0300]	.893	1.000
BN-marg vs BN-base	−0.0123	[−0.0367, .0101]	.366	1.000

*Difference in deterioration of worst-group recall; negative favours BN-marg. The paired natural-test difference between BN-marg and LR-imp was +0.018 in worst-group recall (95% CI $[-0.024, 0.063]$), −0.003 in ΔEOpp ($[-0.073, 0.076]$) and −0.004 in recall ($[-0.018, 0.011]$). Only four race×gender cells have at least 30 positives (181–928 each), leaving wide uncertainty.

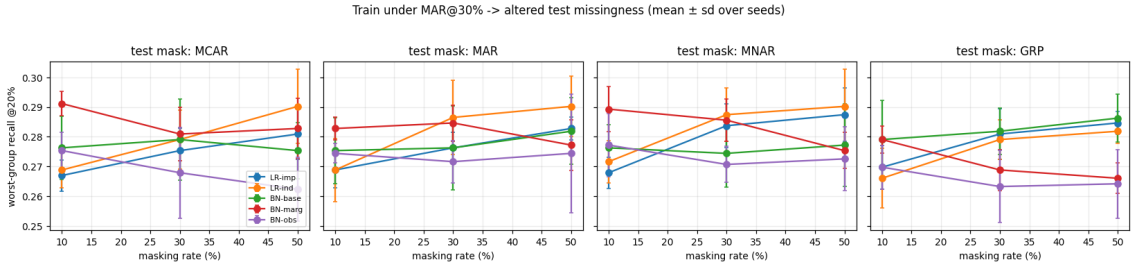


Figure 2: Worst-group recall@20% across altered missingness scenarios (mean ± SD over five seeds). No consistent model ranking emerges.

Across the shift grid, recall@20% remains approximately 0.31–0.33 for all models, consistent with weak predictive signal. PAT does not reliably improve fairness: under group-dependent missingness it lowers LR-ind worst-group recall from 0.279 to 0.228 and increases ΔEOpp from 0.083 to 0.139. The group-aware EO baseline also worsens the natural-data gap (0.091 to 0.146). Under MNAR, BN conditional imputation of masked A1C is less accurate than mode imputation (0.618 vs. 0.693); BN-obs also extrapolates poorly for an unseen missingness pattern. Complete-case LR retains only 3,351 test encounters (16.9%) and has the lowest worst-group recall (0.2059) and largest gap (0.2077), although this is an exploratory comparison on a different population.

4 Limitations, ethics and conclusion

AUROC is about 0.62 for all models. Small race groups are underpowered; MAR for unobserved values is not testable, and MNAR/group-dependent masks are synthetic rather than measured hospital mechanisms. The data cover 1999–2008 and lack hospital identifiers, so results do not establish temporal, hospital-level or prospective performance. The label captures only readmissions recorded in this system; deaths, outside-hospital readmissions and loss to follow-up are not observed. A1C missingness reflects care and documentation as well as patient status. Risk ranking could therefore penalise poorly documented patients; any real workflow would require human review for incomplete records. These models are not for clinical use.

Conclusion. Explicit availability modelling maintained utility within the pre-registered margin but showed no demonstrable improvement in worst-group recall under missingness shift. PAT did not improve the primary fairness gap. The clearest practical warning is against complete-case analysis. Larger, more representative cohorts or stronger predictors are needed to detect effects smaller than roughly 0.03.

Reproducibility. Run `AFGML.ipynb` top-to-bottom (set `QUICK=True` for a smoke test); inputs are UCI dataset 296 files `diabetic_data.csv` and `IDS_mapping.csv`. Split seed is 42; run configuration, versions and data hash are recorded in the evidence bundle. The notebook checks inference against enumeration, `pgmpy` VE and clique-tree inference.

Contributions and AI use. Student 1: Rutvik Gokhale; role/cross-review: [complete]. Student 2: Harshad K; role/cross-review: [complete]. AI assistance: Claude assisted with drafting the original report/slides; [complete any additional tool use].

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